

COMP 3270 – Intro to Algorithms

Homework 1: Induction practice

Due 08/31/25

50 points

1. Find a counterexample to show that for any positive integers a and b , a^b is not always greater than $\max(a, b)$.

2. Use induction to prove that if a_n is a sequence such that $a_0 = 0$ and $a_n = 2a_{n-1} + 2^n$ for $n > 0$, then $a_n = n2^n$ for all $n \geq 0$.

3. Number of Handshakes in a Meeting

Scenario: In a meeting with n people, each person shakes hands with every other person exactly once. You want to find the total number of handshakes.

Claim: The total number of handshakes in a meeting with n people is given by:

$$H(n) = \frac{n(n-1)}{2}$$

4. Prove by induction that for $n \geq 1$,

$$a. \sum_{i=1}^n 3i^2 - 3i + 1 = n^3$$

$$b. 1 + 4 + 7 + \dots + (3n - 2) = \frac{n(3n - 1)}{2}$$

5. Restaurant table reservation system.

Problem statement: Inefficient reservation management in a restaurant leads to double bookings, unbalanced table occupancy, and customer dissatisfaction.

Problem-Solving Approach: Define the problem, Analyze the requirements, identify patterns, design algorithm.

Discuss all the steps by giving the steps as discussed in class. **“Writing program is not required.”**